METRIC
MIL-PRF-62382C
4 January 2011
SUPERSEDING
MIL-PRF-62382B(AT)
19 October 1995

PERFORMANCE SPECIFICATION

VALVE, HYDRAULIC, PRESSURE RELIEF (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of in-line hydraulic pressure relief valve used in a military combat vehicle to control system differential pressure at a predetermined level (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

Comments, suggestions, or questions on this document should be addressed to U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to DAMI_STANDARDIZATION@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

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FEDERAL STANDARDS

FED-STD-595/24533 - Seafoam Green

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-5606 - Hydraulic Fluid, Petroleum Base: Aircraft, Missile and Ordnance
MIL-PRF-6083 - Hydraulic Fluid, Petroleum Base: For Preservation and Operation

(Copies of these documents are available from https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

U.S. ARMY TACOM DRAWINGS

12265558 - Valve, Pressure Relief

(Copies of these documents are available from DAMI_STANDARDIZATION@conus.army.mil or U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted)

(Copies of this document are available from www.asq.org or American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

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ASTM INTERNATIONAL

ASTM B117 - Standard Test Method of Salt Spray (Fog) Testing

(Copies of these documents are available from www.astm.org or ASTM International, P.O. Box C700, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a first article sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for hydraulic valves. The materials shall be capable of meeting all of the operational and environmental requirements specified herein (see 4.5.1).

3.2.1 Age-sensitive materials. The use of age-sensitive materials shall be as specified in the contract or order (see 6.2). The maximum age of the valve poppet seal (see 3.2.5) shall not exceed 40 quarters at time of acceptance of the valve by the Government.

3.2.2 Fungus. Only materials which are fungus non-nutrient shall be used.

3.2.3 Hydraulic fluids. The valve shall be compatible with hydraulic fluids conforming to MIL-PRF-5606 and MIL-PRF-6083.

3.2.4 Metals. All metals used in the construction of the valves except metal which is in constant contact with the hydraulic fluid, shall be of a corrosion-resistant type or shall be suitably protected to resist corrosion during the normal service life of the valves. The use of dissimilar metals, especially brass, copper, or steel in intimate metal to metal contact, shall be avoided.

3.2.5 Poppet seal. The valve poppet seal shall be of elastomer material to minimize leakage under shock and vibration conditions.

3.2.6 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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3.3 Design and construction. The valve shall conform to the manufacturer's design and construction for hydraulic pressure relief valves except for outline dimensions, and porting hole dimensions and locations which shall conform to Drawing 12265558 (see 4.5.2).

3.3.1 End cap. The threaded end cap (outlet port) of the valve shall be safety-lockwired with 0.51 millimeter (mm) (0.020 inch) diameter corrosion resistant steel wire in accordance with Drawing 12265558.

3.4 Performance.

3.4.1 Proof pressure. The valve shall withstand an internal pressure of 31 000 kilopascals (kPa) (4500 pounds per square inch gage [psig]) without leaking externally. The valve shall not be damaged and shall subsequently meet the requirements of 3.4.3 and 3.4.4 (see 4.5.4.1).

3.4.2 Burst pressure. The valve shall withstand an internal pressure of 52 000 kPa (7500 psig) without rupture of internal or external parts. The valves need not be operable after this exposure (see 4.5.4.2).

3.4.3 Function. Unless otherwise specified herein, the valve shall function as specified below with hydraulic pressure applied to the inlet port.

3.4.3.1 Cracking pressure. The valve poppet shall crack open when the differential pressure across the valve is not more than 140 kPa (20 psi). The flow after cracking shall be not less than 1.9 liters per minute (L/min) (0.5 gallons per minute [gal/min]) at 140±7 kPa (20±1 pounds per square inch differential [psid]) (see 4.5.4.3).

3.4.3.2 Rated flow and pressure drop. With fluid flow through the valve of 20 L/min (5 gal/min), the pressure drop through the valve shall not be greater than 100 kPa (15 psid) (see 4.5.4.4).

3.4.3.3 Reseat pressure. After flow is established through the valve, the poppet shall reseat when the differential pressure is not less than 70 kPa (10 psid). No flow shall be permitted in the reverse direction (see 4.5.4.5).

3.4.4 Low pressure leakage. There shall be no evidence of leakage through the valve when a pressure of 35 kPa (5 psig) is applied at the inlet port; there shall be no evidence of leakage when 200 kPa (30 psig) is applied at the outlet port (see 4.5.4.6).

3.4.5 Environmental. The valve shall meet the requirements of 3.4.1 and shall operate as specified in 3.4.3 and 3.4.4 under the following environmental conditions.

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3.4.5.1 High temperature. The valve shall operate after exposure to an ambient temperature of +70 degrees Celsius (°C) (+160 degrees Fahrenheit [°F]) and during and after exposure to +60 °C (+140 °F) (see 4.5.5.1).

3.4.5.2 Low temperature. The valve shall operate after exposure to an ambient temperature of -54 °C (-65 °F), and during and after exposure to -32 °C (-25 °F) (see 4.5.5.2).

3.4.5.3 Fluid temperature. The valve shall operate when exposed to hydraulic fluid temperatures ranging from +70 °C (+160 °F) to -32 °C (-25 °F) (see 4.5.5.3).

3.4.5.4 Salt fog. The valve shall operate after exposure to a salt-fog environment (see 4.5.5.4).

3.4.5.5 Shock resistance. The valve shall operate after exposure to a shock impulse of 25 gravity units (g) for 11 +0/-1 milliseconds (ms), half-sine wave, applied along three mutually perpendicular axes (see 4.5.5.5).

3.4.5.6 Vibration resistance. The valve shall operate after exposure to the following vibration conditions (see 4.5.5.6).

a. Simple harmonic motion.
b. Motion applied along three mutually perpendicular axes.
c. Peak acceleration of 10g at all frequencies.
d. Frequency range, 20 to 2000 Hertz (Hz).
e. Frequency sweep varied logarithmically.
f. Sweep time, 35±5 minutes.
g. Dwell 5 minutes at each observed resonant frequency.

3.5 Cleanliness. Upon completion of testing, sufficient quantity of fluid shall be retained in the valve to assure proper preservation during extended storage. Ports shall be covered with suitable plugs. Valve internal cleanliness shall conform to the level shown in table I (see 4.6).

TABLE I. Valve internal cleanliness.

Maximum size of surface particles (micrometers)	Maximum quantity of particles
25	7450
50	1020
100	95
250	2

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3.6 Painting. The exterior surfaces of the valve shall be prepared, primed or treated, and painted as suitable for the material of the valve body (see 4.5.2). Color of finish shall be seafoam green, #24533 of FED-STD-595.

3.7 Marking. The valve shall be marked with the National Stock Number (NSN) and the manufacturer's name and part number. Marking shall be permanent and legible (see 4.5.2).

3.8 Workmanship. Workmanship shall be of the quality as to assure that valves furnished under this specification are free of defects that compromise, limit, or reduce performance in intended use (see 4.5.2).

4. VERIFICATION

4.1 Classification of inspection. The inspection conditions specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on preproduction or initial production samples as specified when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.4 (see table II) and the tests of 4.5.4.1 through 4.6 (see table III).

TABLE II. Classification of defects.

Category	Defect	Method of examination
Major:		
101	Dimensions affecting interchangeability, not within tolerance (see 3.3).	SIE 1/
102	End cap safety-lockwiring, not to requirements (see 3.3.1).	Visual
103	Identification marking, improper (see 3.7).	Visual
104	Faulty workmanship affecting performance (see 3.8).	SIE
Minor:		
201	Dimensions not affecting interchangeability, not within tolerance (see 3.3).	SIE and Visual
202	Painting, improper (see 3.6).	Visual
203	Faulty workmanship affecting appearance (see 3.8).	Visual

1/ SIE = Standard Inspection Equipment.

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TABLE III. Classification of inspections.

Title	Requirement	Inspection	First article	Conformance tests
Proof pressure	3.4.1	4.5.4.1	X	X
Burst pressure	3.4.2	4.5.4.2	X	
Cracking pressure	3.4.3.1	4.5.4.3	X	X
Rated flow and pressure drop	3.4.3.2	4.5.4.4	X	X
Reseat pressure	3.4.3.3	4.5.4.5	X	X
Low pressure leakage	3.4.4	4.5.4.6	X	X
High temperature	3.4.5.1	4.5.5.1	X	
Low temperature	3.4.5.2	4.5.5.2	X	
Fluid temperatures	3.4.5.3	4.5.5.3	X	
Salt fog	3.4.5.4	4.5.5.4	X	
Shock resistance	3.4.5.5	4.5.5.5	X	
Vibration resistance	3.4.5.6	4.5.5.6	X	
Cleanliness	3.5	4.6	X	X

4.2.1 Preproduction inspection. When specified (see 6.2), first article preproduction inspection shall be performed on two valves. This inspection shall include the examinations of 4.4 (see table II) and the tests of 4.5.4.1 through 4.6 (see table III).

4.2.2 Initial production inspection. Unless otherwise specified (see 6.2), first article initial production inspection shall be performed on two valves. This examination shall include the examinations of 4.4 (see table II) and the tests of 4.5.4.1 through 4.6 (see table III).

4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.4 (see table II) and the tests of 4.5.4.1 through 4.5.4.6 and 4.6 (see table III).

4.4 Examination.

4.4.1 Sampling. Samples from an inspection lot for conformance inspection shall be selected in accordance with ANSI/ASQ Z1.4. Any redesign or modification of the contractor's standard to comply with specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements of requirements 3.3, 3.3.1, 3.6, 3.7 and 3.8 as listed in table II. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Method of inspection.

4.5.1 Materials. Conformance to 3.2 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records,

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processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.5.2 Defects. Conformance to 3.3, 3.3.1, 3.6, 3.7 and 3.8 shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with SIE.

4.5.3 Test conditions. Unless otherwise specified (see 6.2), ambient air temperature shall be 23±10 °C (73±18 °F), barometric pressure shall be 725 +50/-75 mm mercury (Hg) (28.5 +2/-3 inches Hg), and relative humidity shall be 50±30 percent. Unless otherwise specified herein, the valve shall meet all specified performance requirements in a hydraulic system, utilizing hydraulic fluid conforming to MIL-PRF-5606 or MIL-PRF-6083 or the commercial equivalents under the following conditions:

a. Hydraulic fluid temperature of 38±5 °C (100±10 °F).
b. Hydraulic fluid operating pressure of 20 700±700 kPa (3000±100 psi).

4.5.4 Performance.

4.5.4.1 Proof pressure. To determine conformance to 3.4.1, the valve shall be subjected to a static pressure test of 31 000 kPa (4500 psi) for a duration of 2 minutes. There shall be no external leakage. The test shall be performed with pressure applied to the inlet port with the outlet port plugged.

4.5.4.2 Burst pressure. To determine conformance to 3.4.2, the valve shall be subjected to a static pressure test of 52 000 kPa (7500 psig) for a duration of 2 minutes. Pressure shall be applied to the inlet port, with the outlet port plugged. External or internal parts of the valve shall not rupture. This is a destructive test, so test valves shall be indelibly marked “DO NOT USE” and discarded.

4.5.4.3 Cracking pressure. To determine conformance to 3.4.3.1, pressure shall be applied slowly to the valve inlet until the valve poppet cracks open. The pressure differential at the instant of valve opening shall not be more than 140 kPa (20 psid). After opening, the flow shall be not less than 1.9 L/min (0.5 gal/min) at 140±7 kPa (20±1 psid).

4.5.4.4 Rated flow and pressure drop. To determine conformance to 3.4.3.2, pressure shall be applied to the valve inlet and the flow shall be adjusted to 20 L/min (5 gal/min). The pressure drop through the valve shall not be more than 100 kPa (15 psi).

4.5.4.5 Reseat pressure. To determine conformance to 3.4.3.3, a pressure of 172 kPa (25 psi) shall be applied to the valve inlet and flow through the valve verified. Then the pressure shall be gradually decreased until the valve closes. The differential pressure at the instant of valve closing shall be not less than 70 kPa (10 psi).

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4.5.4.6 Low pressure leakage. To determine conformance to 3.4.4, alternately apply a pressure of 35 kPa (5 psi) to the valve inlet, then 200 kPa (30 psig) to the outlet port. There shall be no evidence of leakage through the valve.

4.5.5 Environmental.

4.5.5.1 High temperature. To determine conformance to 3.4.5.1, the valve shall be heated in a chamber until its temperature stabilizes at +70 °C (+160 °F). Remove the valve from the chamber and allow it to cool to ambient room temperature. The valve shall meet the applicable requirement of 3.4.5.1. Next, heat the valve in a chamber until its temperature stabilizes at +60 °C (+140 °F). The valve shall meet the applicable requirement of 3.4.5.1 at the stabilized temperature and after it is cooled to ambient room temperature.

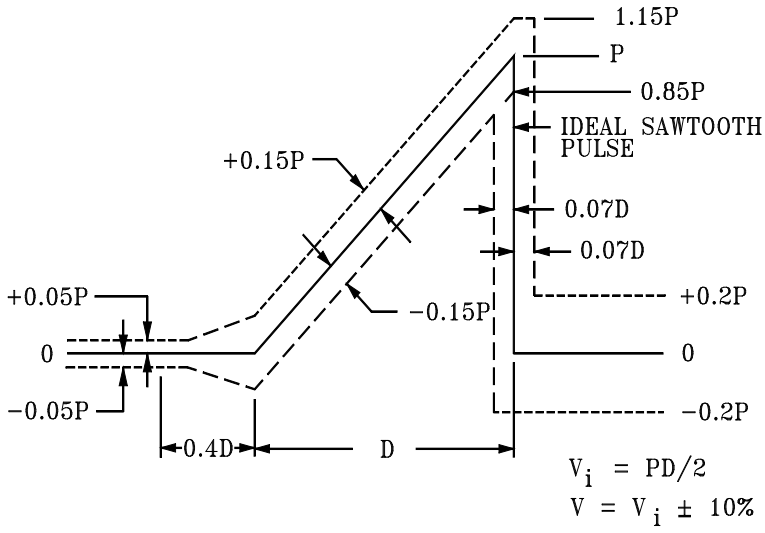
4.5.5.2 Low temperature. To determine conformance to 3.4.5.2, the valve shall be cooled in a chamber until its temperature stabilizes at -54 °C (-65 °F). Remove the valve from the chamber and allow it to warm to ambient room temperature. The valve shall meet the applicable requirement of 3.4.5.2. Next, cool the valve in a chamber until its temperature stabilizes at -32 °C (-25 °F). The valve shall meet the applicable requirement of 3.4.5.2 at the stabilized temperature and after it is warmed to ambient room temperature.

4.5.5.3 Fluid temperature. Conformance to 3.4.5.3 shall be determined during testing in accordance with 4.5.5.1 and 4.5.5.2.

4.5.5.4 Salt fog. To determine conformance to 3.4.5.4, the valve shall be exposed to a salt fog environment in accordance with ASTM B117.

4.5.5.5 Shock resistance. To determine conformance to 3.4.5.5, the valve shall be subjected to a half sine shock pulse as shown in figure 1.

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NOTE:
The oscillogram should include a time about 3D long with the pulse located approximately in the center. The integration to determine velocity change should extend from .4D before the pulse to .1D beyond the pulse. The acceleration amplitude of the ideal half sine pulse is A and its duration is D. Any measured acceleration pulse which can be contained between the broken line boundaries is a nominal half sine pulse of nominal amplitude A and nominal duration D. The velocity-change associated with the measured acceleration pulse is V.

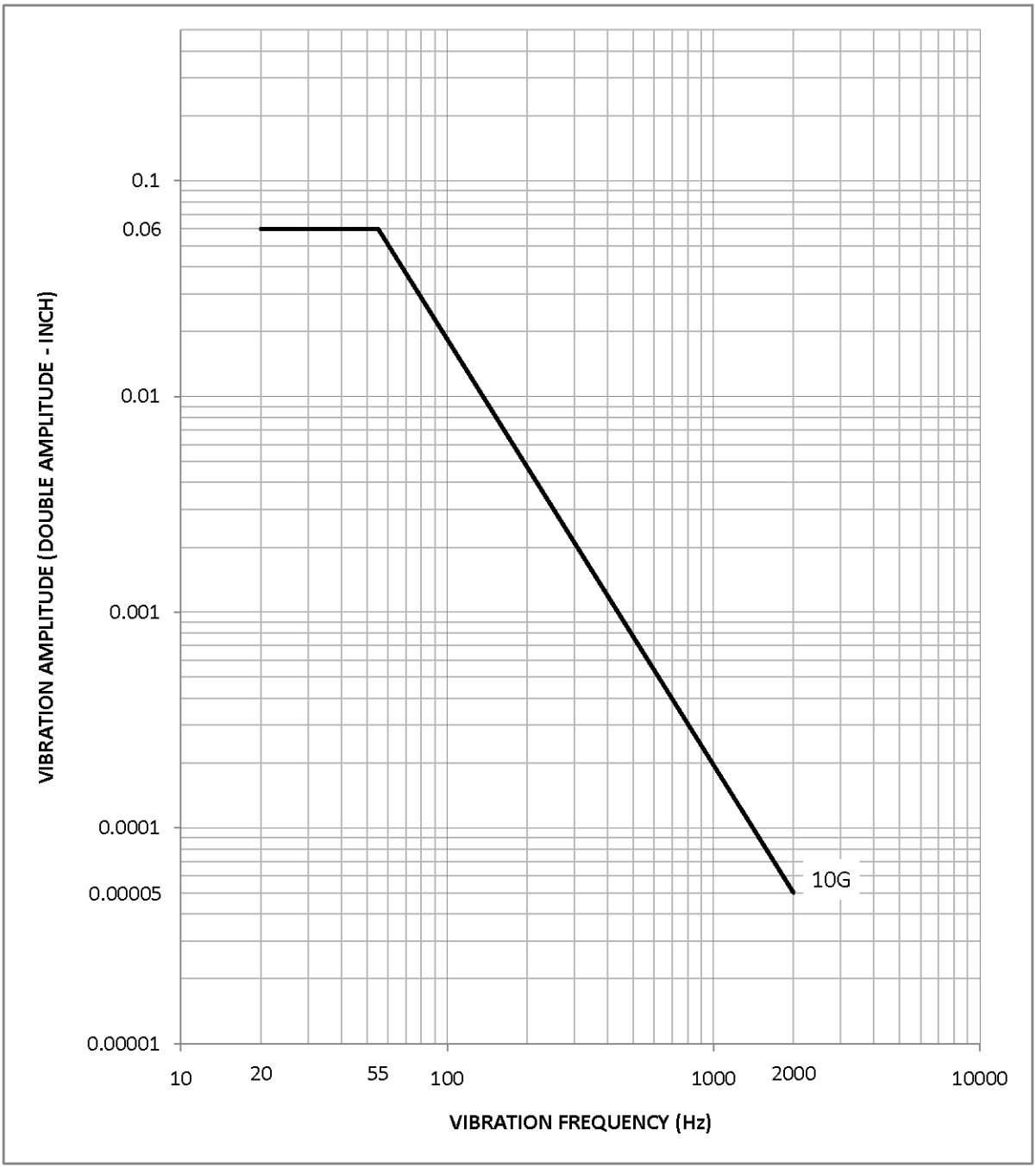
FIGURE 1. Tolerances for half sine shock pulse.

4.5.5.6 Vibration resistance. To determine conformance to 3.4.5.6, the valve shall be subjected to a high frequency vibration as shown in figure 2.

4.6 Cleanliness. To determine conformance to 3.5, the following shall be performed:

a. Prior to any performance testing, the internal cleanliness of the valve shall be established by providing operating pressure at the level specified in 4.5.3 and then establishing a flow between the two ports. Maintain the flow until the effluent meets the cleanliness of table I.
b. At the conclusion of quality conformance testing, re-establish the assemblies internal cleanliness by performing the same flushing and effluent check. Once the required cleanliness has been achieved, drain the effluent and seal the assembly with a non-contaminating, tamper proof seal, adequately tagged and identified, before submitting it to preparation for delivery or storage processing.

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$G = 0.0512f^2DA$ (f - frequency in hertz, DA = double amplitude in mm)

FIGURE 2. Vibration - test curve.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The valve, furnished under this specification, is a component part of the hydraulic control system of the M901 Improved Tow Vehicle (ITV) launcher and the M981 Fire Support Team Vehicle (FISTV) targeting head.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. If first article samples are not required (see 3.1).
d. Instructions or details on the use of age-sensitive materials (see 3.2.1).
e. If preproduction inspection is required (see 4.2.1).
f. If initial production inspection is not required (see 4.2.2).
g. Test conditions if other than as specified (see 4.5.3).
h. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Hydraulic fluid

6.4 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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